光的偏振（Polarization of light）

§1　光的偏振状态

§2　偏振片的起偏和检偏　马吕斯定律

§3　反射和折射光的偏振　散射光的偏振

§4　光在晶体中的传播　双折射

§5　椭圆偏振光和圆偏振光

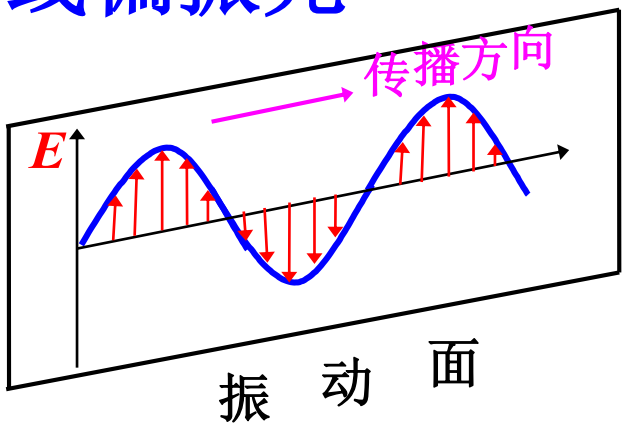
§6　偏振光的干涉

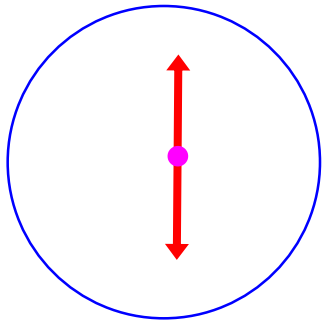
§7　人工双折射

§8　*旋光现象

§1 光的偏振状态

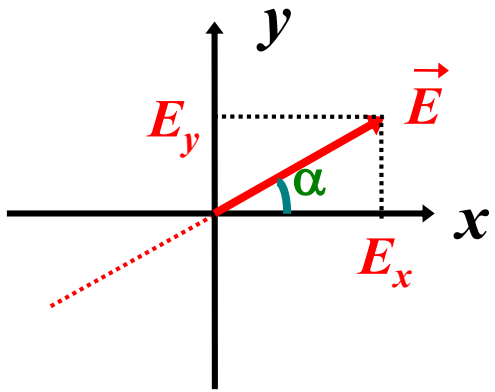
一. 线偏振光





面对光的传播方向看

线偏振光可沿两个相互垂直的方向分解

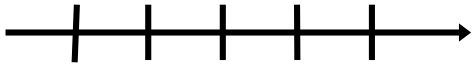


$$E_x = E\cos\alpha$$
$$E_y = E\sin\alpha$$

相位相同

线偏振光的表示法：

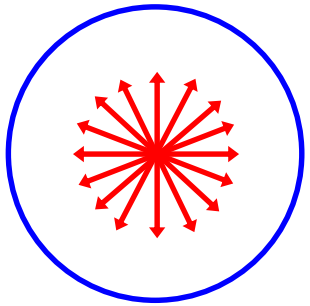
光振动垂直板面



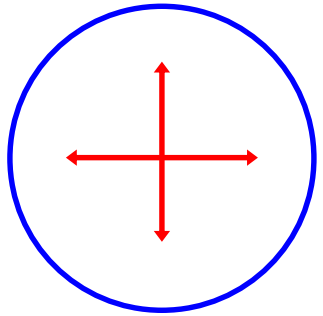
光振动平行板面

二. 自然光

- 不相干的线偏振光集合
- 本身就是非偏振光



没有优势方向

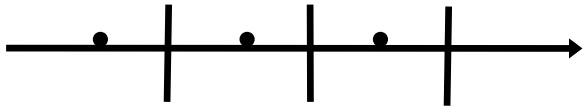


自然光的分解

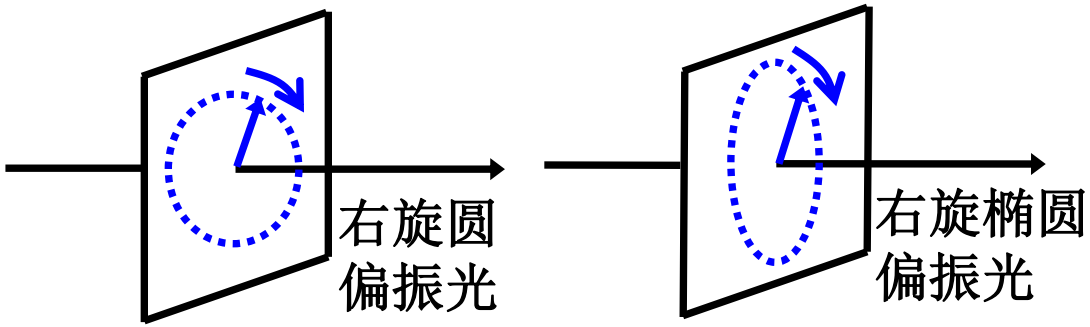
一束自然光可分解为两束振动方向相互垂直的、等幅的、相位差不固定的线偏振光

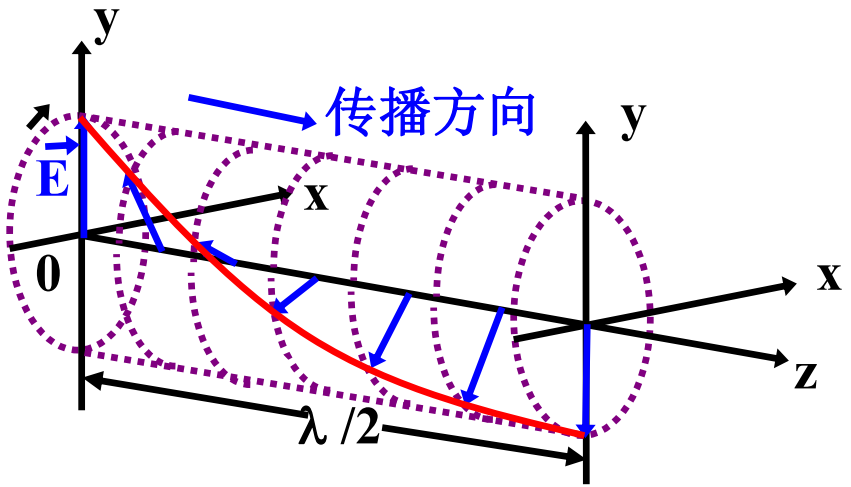
$$\overline{E}_x = \overline{E}_y \qquad I = I_x + I_y$$

自然光的表示法：

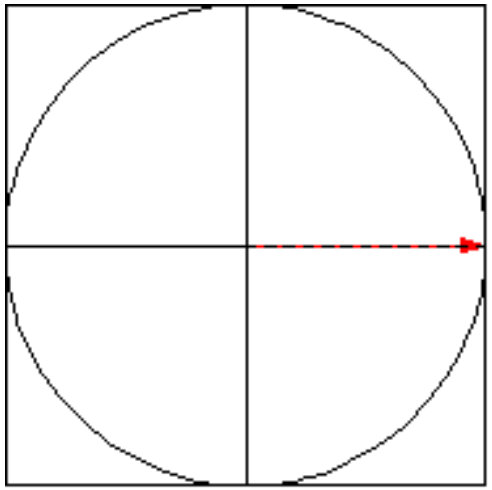


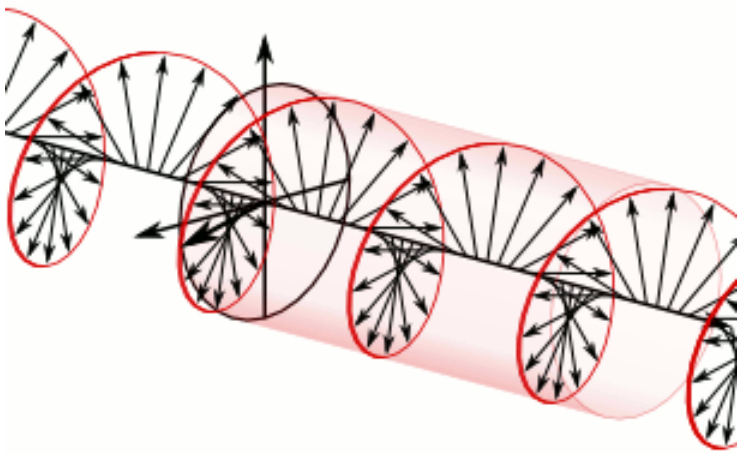
三. 圆偏振光， 椭圆偏振光





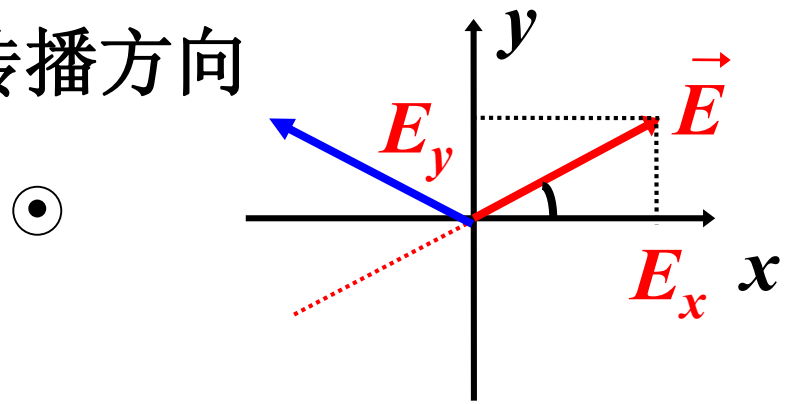
某时刻右旋圆偏振光 $\vec{E}$ 随z的变化





圆或椭圆偏振光可分解为沿两个相互垂直方向的线偏振光

光传播方向



$$E_x = E_1 \cos(\omega t + \varphi_1)$$
$$E_y = E_2 \cos(\omega t + \varphi_2)$$

$E_1 = E_2$ ， $\varphi_1 - \varphi_2 = \pi/2$ 左旋圆 $\varphi_1 - \varphi_2 = 3\pi/2$ 右旋圆

$E_1 \neq E_2$ ， $\varphi_1 - \varphi_2 = \pi/2$ 左旋正椭圆 一般斜椭圆

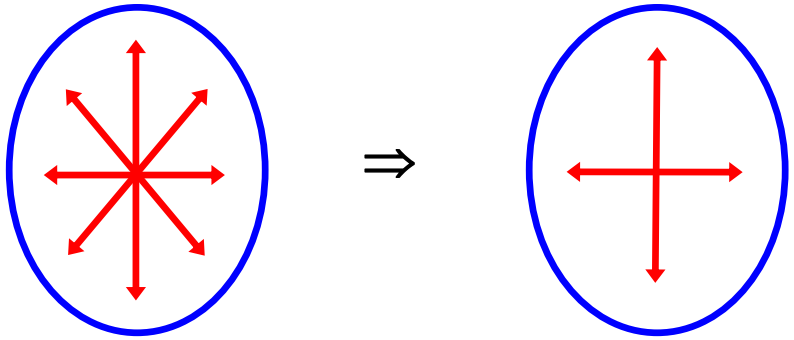
$\varphi_1 - \varphi_2 = 0$ 或 π 线偏振光

$\alpha = \tan^{-1} E_2 / E_1$

或 $\pi - \alpha$

四. 部分偏振光

- 自然光和线偏振光的混合



部分偏振光　　部分偏振光的分解

部分偏振光可分解为两束振动方向相互垂直的、不等幅的、不相干的线偏振光。

这种部分偏振光的表示法：

平行板面的光振动较强　　垂直板面的光振动较强

- 自然光和圆(椭圆)偏振光的混合

五. 偏振度

$$P = \frac{I_p}{I_t} = \frac{I_p}{I_n + I_p}$$

I_t —部分偏振光的总强度

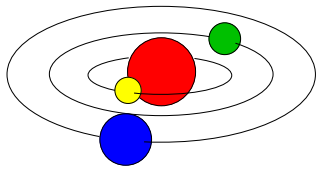
I_n —部分偏振光中包含的自然光的强度

I_p —部分偏振光中包含的完全偏振光的强度

完全偏振光（线、圆、椭圆 ） $P = 1$

自然光（ 非偏振光 ） $P = 0$

部分偏振光 $0 < P < 1$

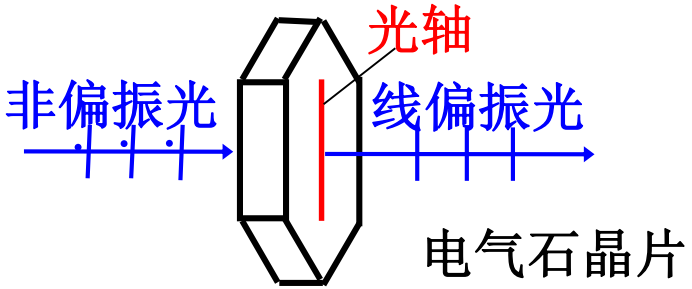


§2 偏振片的起偏和检偏，马吕斯定律

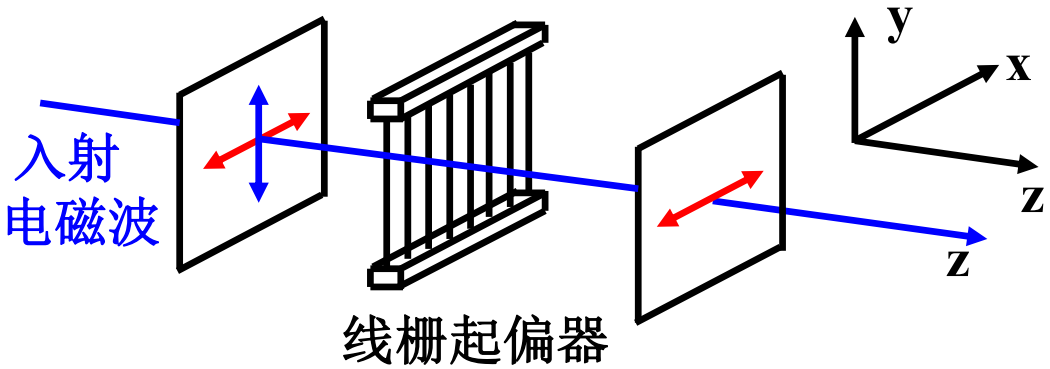
一. 起偏

- 起偏：从自然光获得偏振光
- 起偏器： 起偏的光学器件
- 起偏的原理：利用某种光学的不对称性
- 偏振片

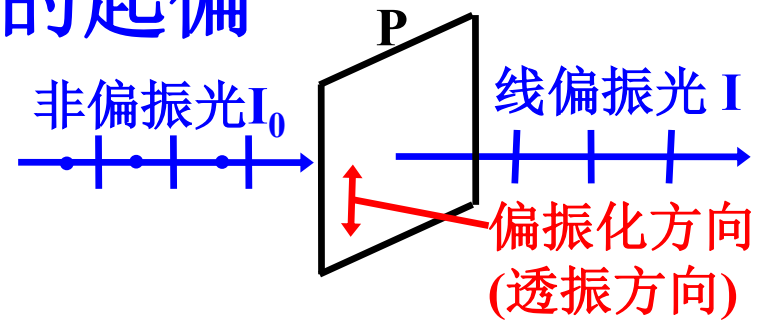
微晶型



分子型

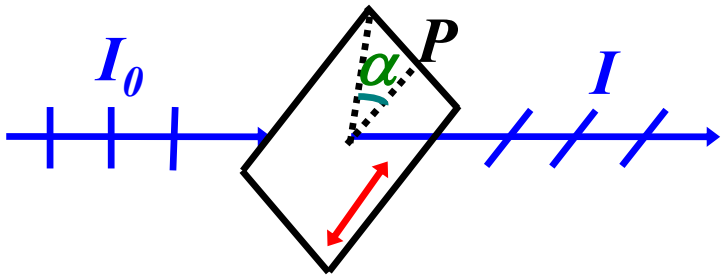


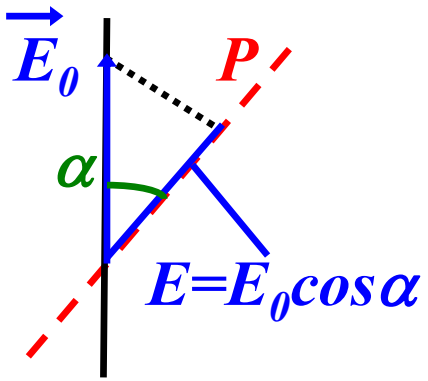
- 偏振片的起偏



$$I = \frac{1}{2} I_0$$

二. 马吕斯定律





$$I_0 \propto E_0^2 \ , \qquad I \propto E^2 = E_0^2 \cos^2 \alpha$$

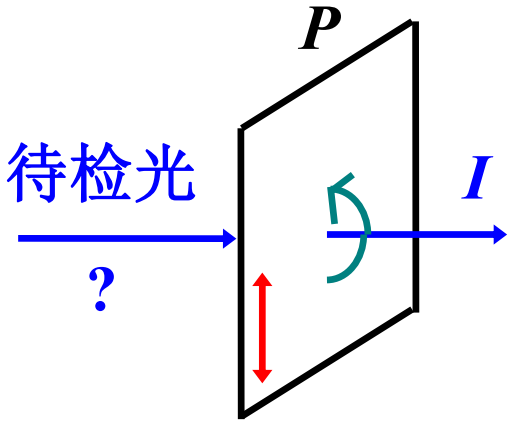
$$I = I_0 \cos^2 \alpha$$ ——马吕斯定律（1809）

$$\alpha = 0, \quad I = I_{\max} = I_0$$

$$\alpha = \frac{\pi}{2}, \quad I = 0$$ ——消光

三. 检偏

用偏振器件分析、检验光的偏振态



通过转动偏振片
观察光强变化

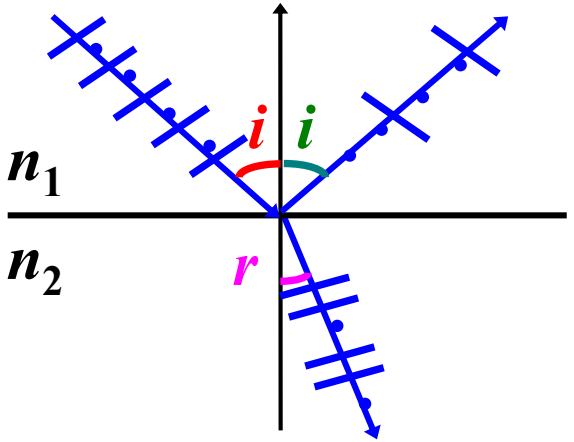
自然光和圆偏振光

部分偏振光 (线偏振光+自然光) 和椭圆偏振光

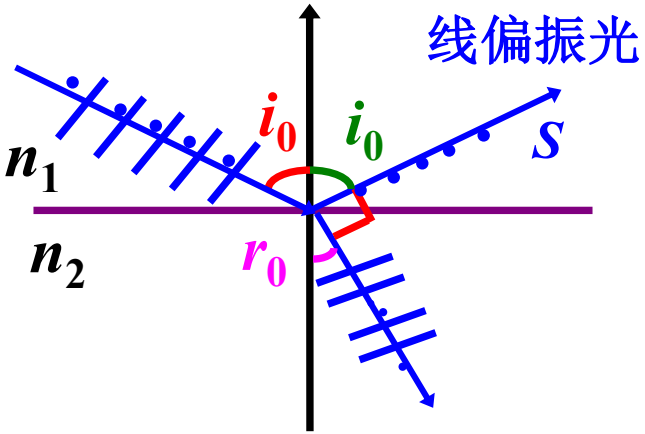
线偏振光

§3 反射和折射光的偏振　散射光的偏振

一. 反射和折射光的偏振



自然光反射和折射后产生部分偏振光



起偏振角

$i = i_0$ 时，反射光只有S分量

i_0 — 布儒斯特角或 起偏角　　$i_0 + r_0 = 90^\circ$

由　$n_1 \sin i_0 = n_2 \sin r_0 = n_2 \cos i_0$

有　$\mathrm{tg}\, i_0 = \dfrac{n_2}{n_1} = n_{21}$　—布儒斯特定律（1812年）

未加偏振片时
拍摄的橱窗

加偏振片时
拍摄的橱窗

(a)

(b)

未加偏振片时

拍摄的橱窗

加了偏振片后

拍摄的橱窗

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若 n_1 =1.00（空气），n_2 =1.50（玻璃），则：

$$\left.\begin{array}{l}\text{空气}\rightarrow\text{玻璃}\quad i_0=\mathrm{tg}^{-1}\dfrac{1.50}{1.00}=56°18\\[2ex]\text{玻璃}\rightarrow\text{空气}\quad i_0=\mathrm{tg}^{-1}\dfrac{1.00}{1.50}=33°42\end{array}\right\}\text{互余}$$

二. 玻璃片堆起偏和检偏

1. 起偏

$$(1-15\%)^{2\times n}$$

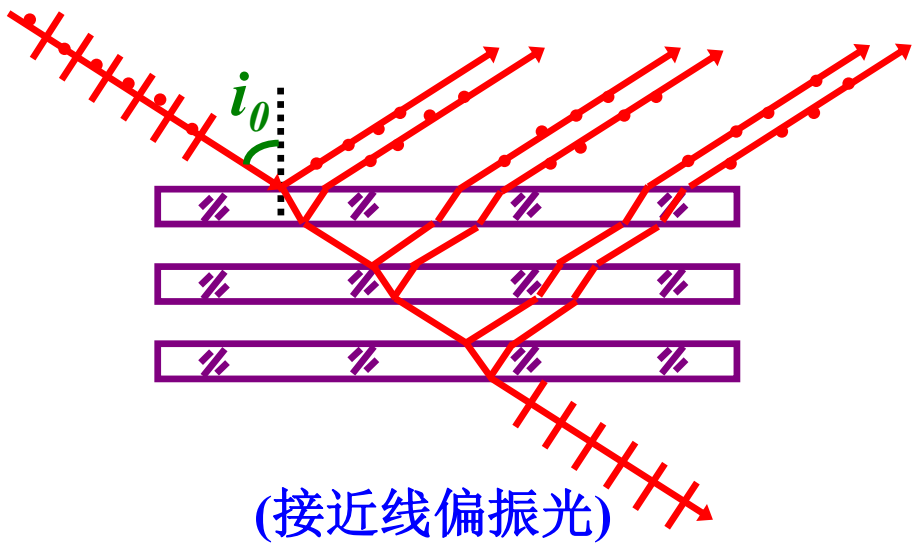
$n=10,\quad 3.9\%\quad P\approx 96\%$

$n=15,\quad 0.76\%,\quad P\approx 99\%$

当 $i=i_0$ 时

自然光从空气→玻璃

$R_S\approx 15\%\qquad R_P=0$

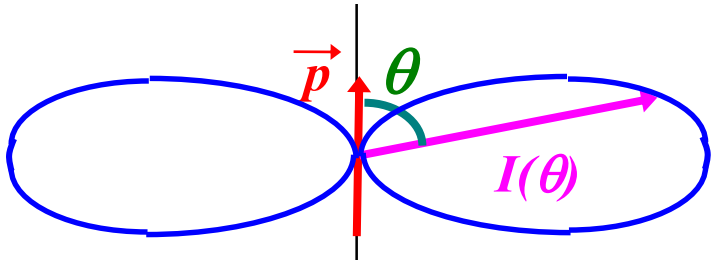


2. 检偏

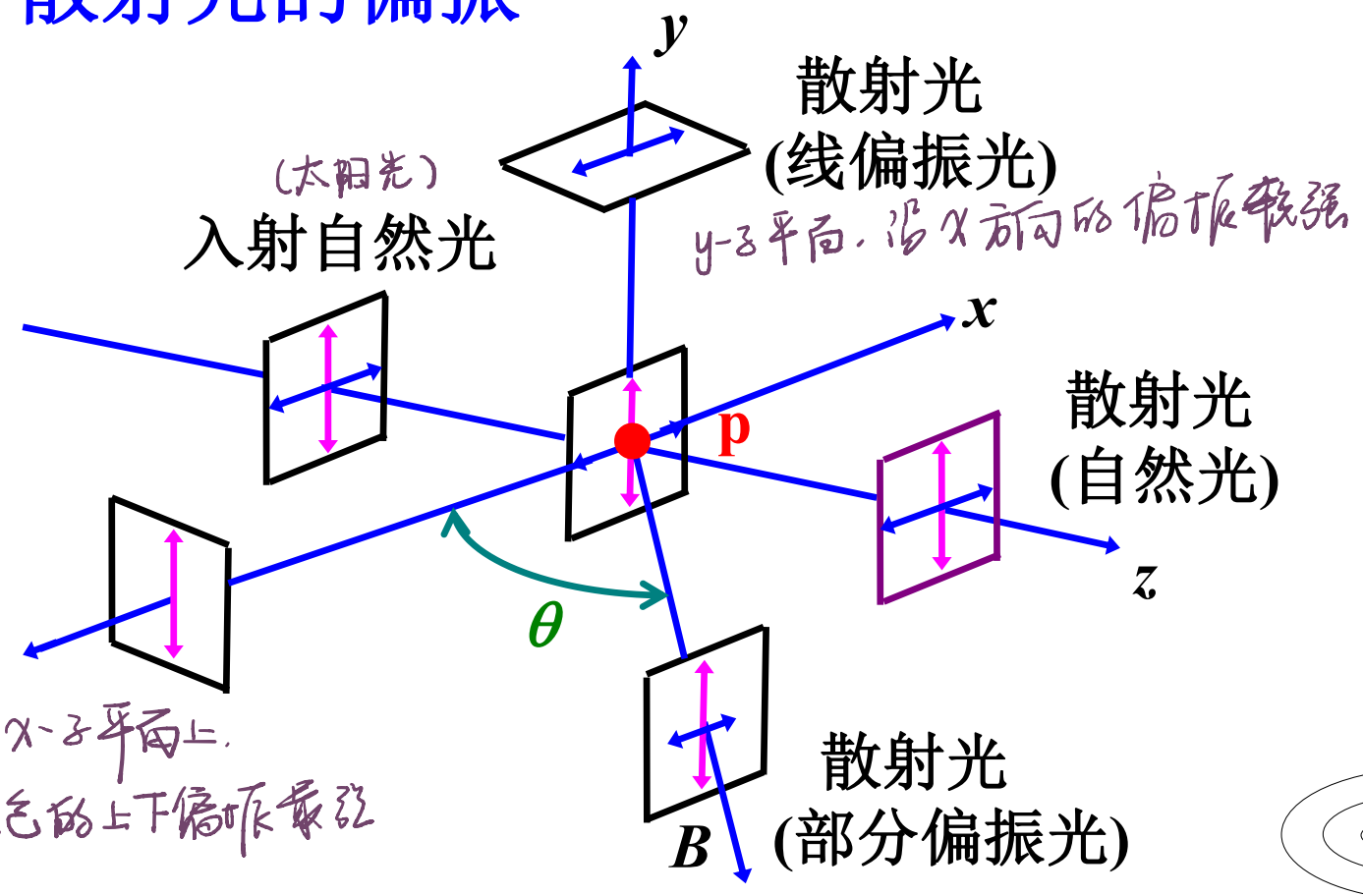
三. 散射光的偏振

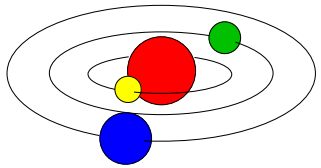
1. 散射光的产生

2. 散射光的偏振



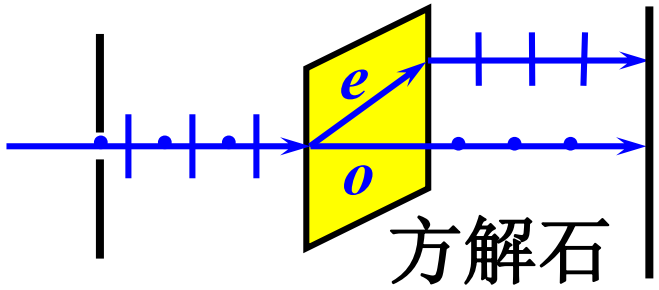
振荡电偶极子电磁
辐射强度的角分布

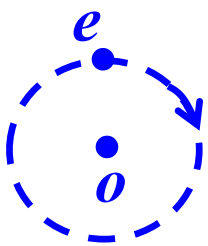




§4　光在晶体中的传播，　双折射

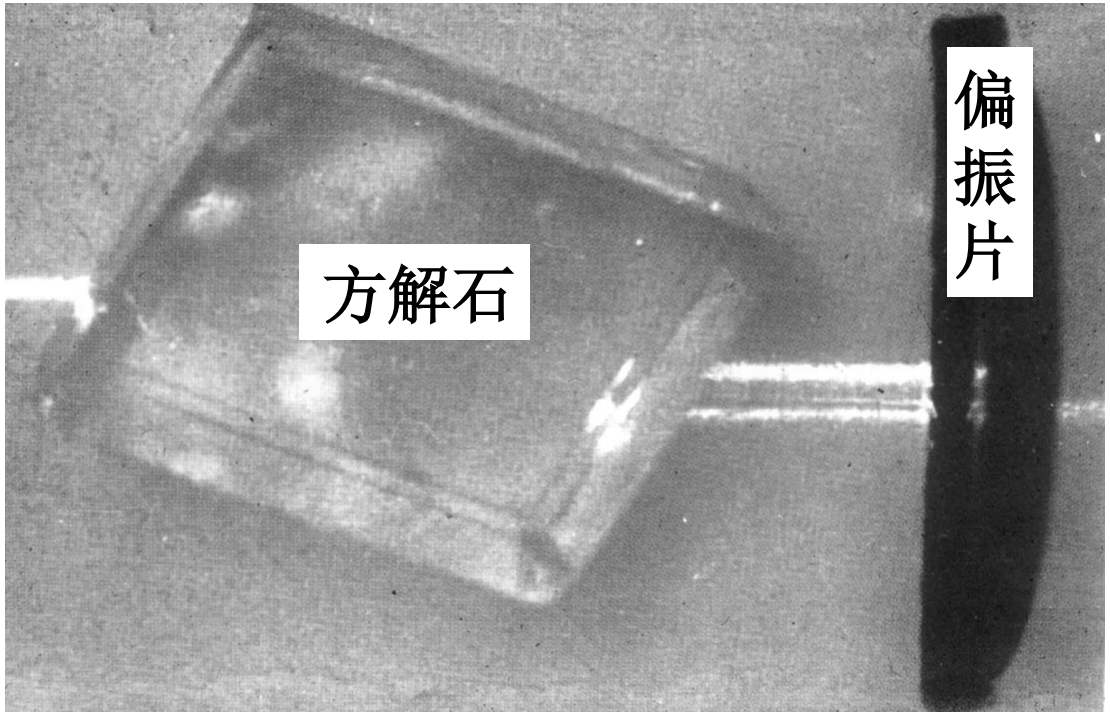
birefringence





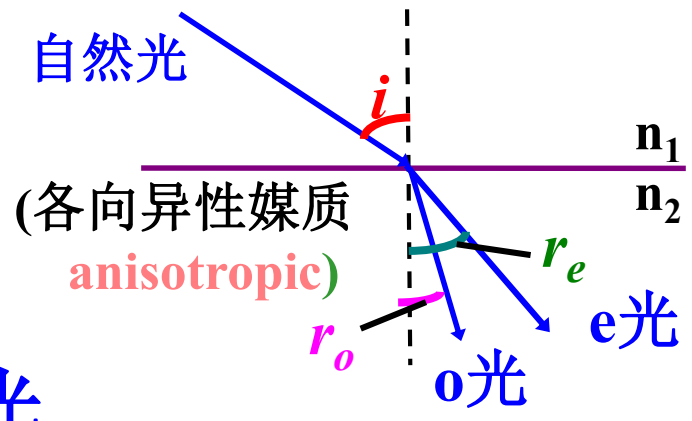
以入射方向为轴旋转方解石

ordinary
extraordinary



双折射的两束光振动方向相互垂直

一. 双折射的概念



1. 双折射

2. 寻常光和非寻常光

*o*光 ： 遵从折射定律 $n_1 \sin i = n_2 \sin r_o$

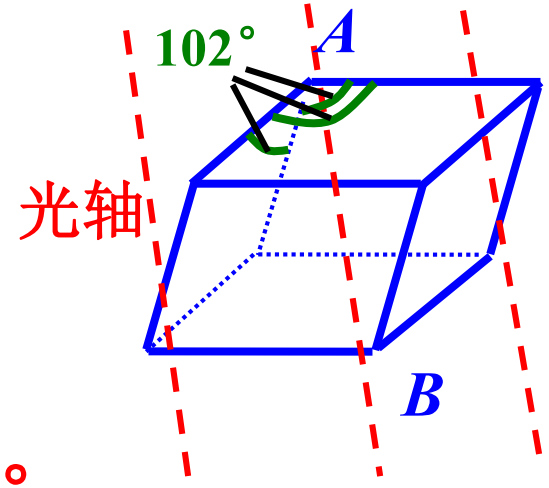
*e*光 ： 一般不遵从折射定律 $\dfrac{\sin i}{\sin r_e} \neq const$

*e*光折射线也不一定在入射面内。

3. 晶体的光轴

当光在晶体内沿某个特殊方向传播时不发生双折射，该方向称为晶体的光轴。

例如，方解石晶体(冰洲石)



- 光轴是一特殊的方向，凡平行于此方向的直线均为光轴。

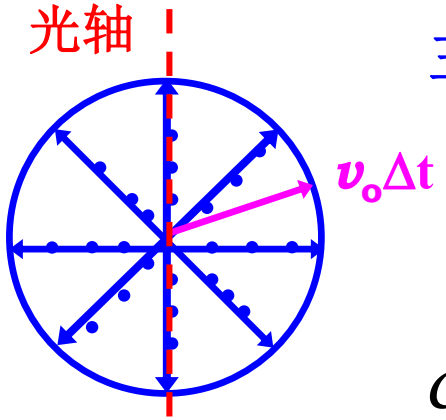
单轴晶体：只有一个光轴的晶体

双轴晶体：有两个光轴的晶体

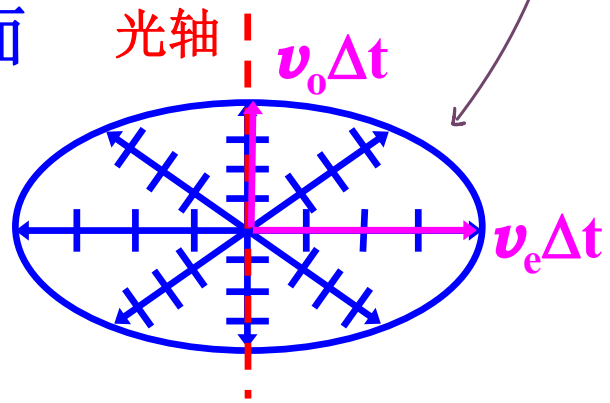
二. 晶体的主折射率，正晶体、负晶体

光矢量振动方向、光线与晶体光轴的夹角，都会影响光的传播速度。

波阵面



主平面



o光垂直于主平面

o光： $n_o = \dfrac{c}{\upsilon_o}$

e光： $\upsilon_o \to n_o$ ， $\upsilon_e \to n_e = \dfrac{c}{\upsilon_e}$

n_o ， n_e 称为晶体（或e光）的主折射率

主平面：光线和光轴组成的平面

*o*光：偏振方向垂直于主平面

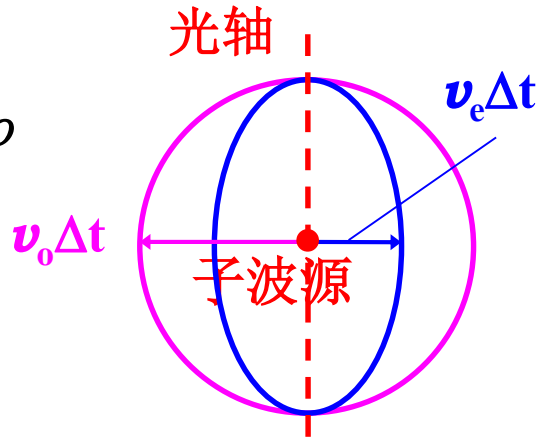
*e*光：偏振方向在主平面内

正晶体 ： $n_e > n_o$

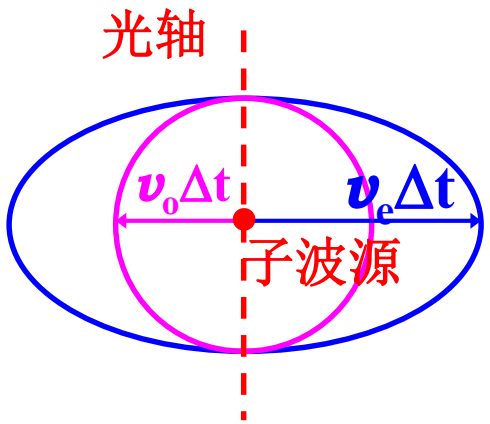
$(\upsilon_e < \upsilon_o)$

负晶体 ： $n_e < n_o$

$(\upsilon_e > \upsilon_o)$



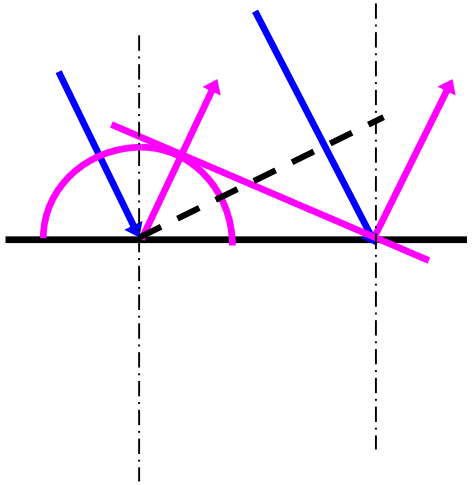
正晶体 ($v_o > v_e$)



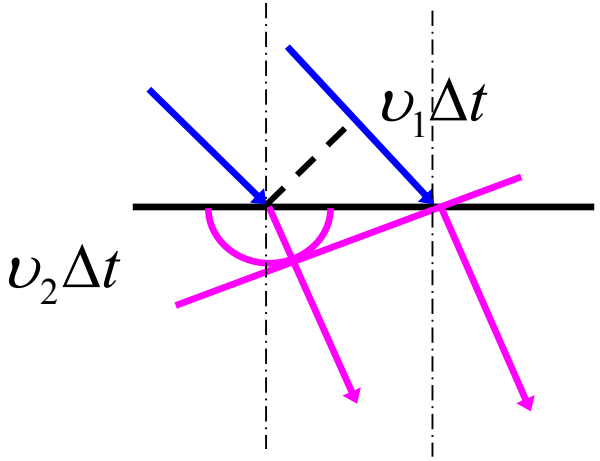
负晶体 ($v_o < v_e$)

三. 单轴晶体中光传播的惠更斯作图法（$\upsilon_e > \upsilon_o$）

复习：反射



折射



$$\frac{\upsilon_1}{\upsilon_2} = \frac{\sin\theta_i}{\sin\theta_r}$$

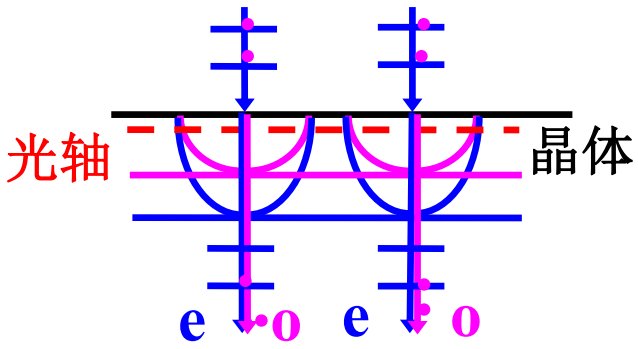
单轴晶体表面折射光的惠更斯作图法（$\upsilon_e > \upsilon_o$）

1. 光轴平行晶体表面，自然光垂直入射

先程：

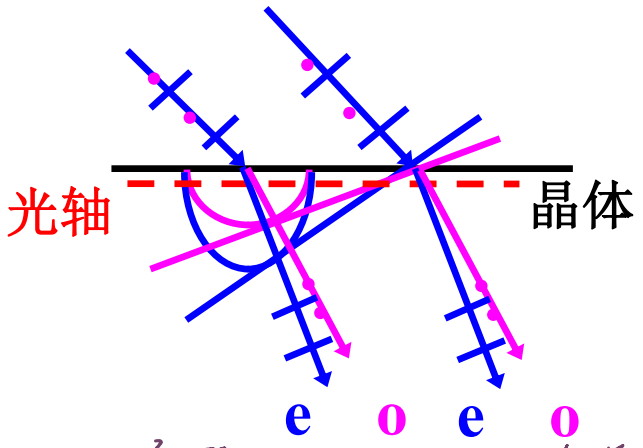
o： $n_o \cdot d$

e： $n_e \cdot d$



o，*e*在方向上虽没分开，但速度上是分开的。

2. 光轴平行晶体表面，自然光垂直入射

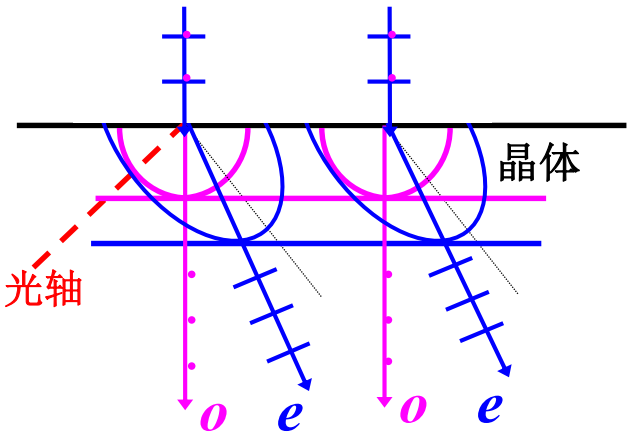


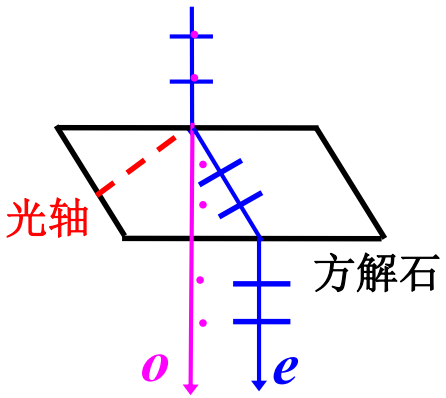
斜入射.

O光满足折射定律

椭球波→e光的折射角度与入射角有关，等相面与传播方向不垂直

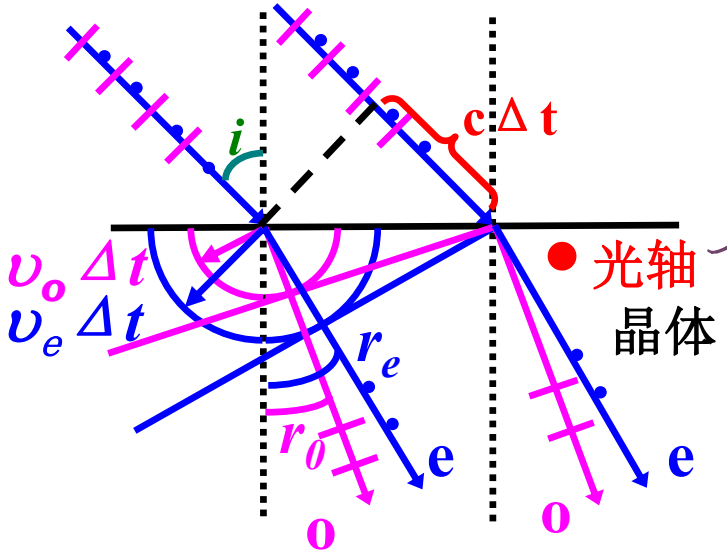
3. 光轴与晶体表面斜交，自然光垂直入射





*e*光的波面与其波射线不垂直

4. 光轴平行晶体表面，且垂直入射面，自然光斜入射



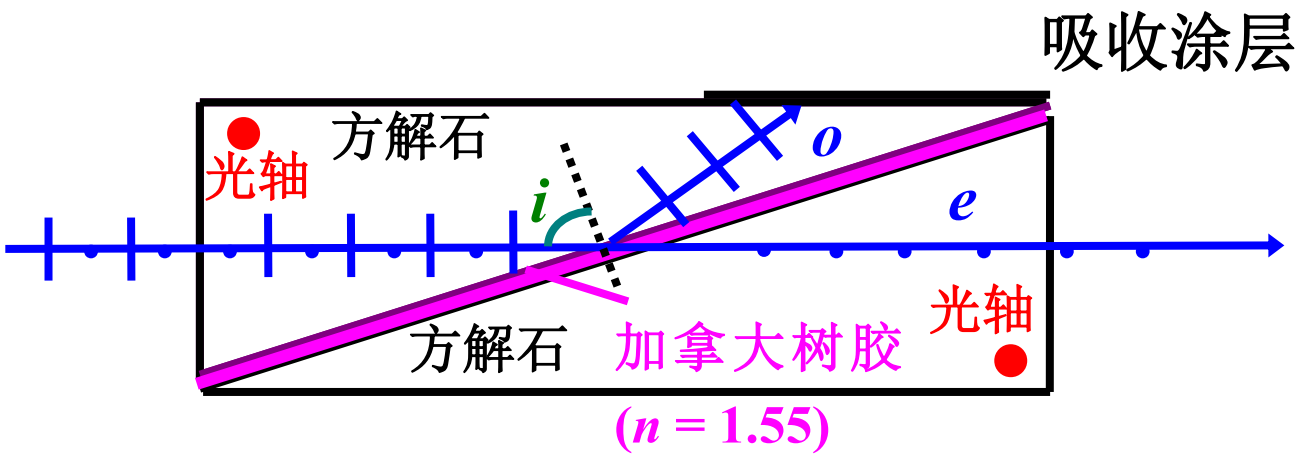
e光的旋转椭球
由侧光轴的椭球
⇒ 沿光轴方向为圆
⇒ 服从折射定律

o，e 折射定律都满足

n_o, n_e

四. 晶体起偏器件

1. 偏振棱镜　偏光镜：格兰—汤姆孙棱镜



光轴的取向对o光、e光可用n_o、n_e

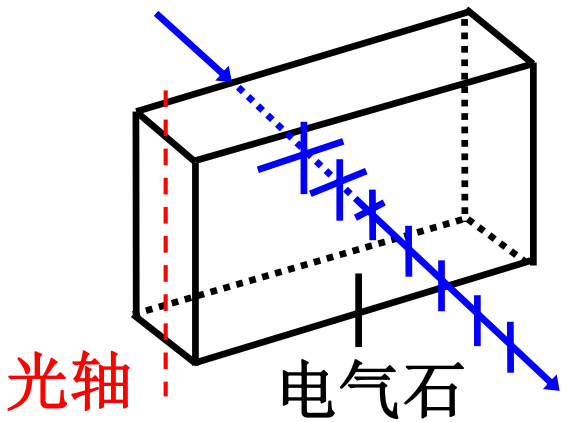
n_o (1.6584) > n (1.55) > n_e(1.4864)

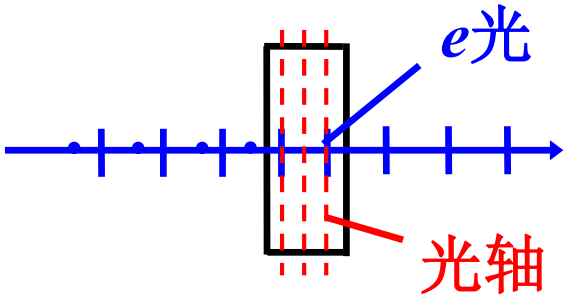
69.2°　　73.5°

2. 晶体的二向色性、晶体偏振器

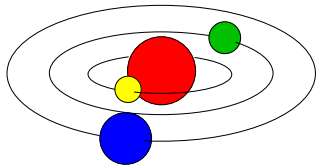
某些晶体对*o*光和*e*光的吸收有很大差异，这叫晶体的二向色性

电气石





可产生线偏振光

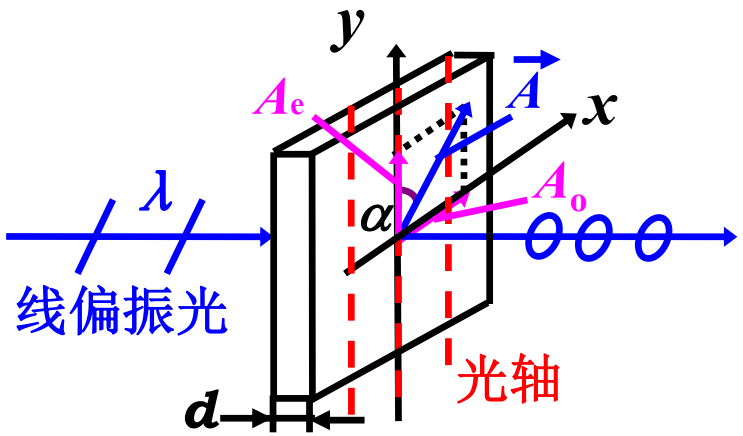


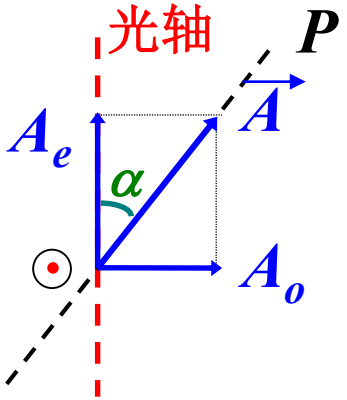
§5　椭圆偏振光和圆偏振光

一．圆和椭圆偏振光的起偏

1．晶片—相位延迟片

晶片是光轴平行表面的晶体薄片。





通过厚为d 的晶片，o、e 光产生相位差：

$$\varphi_e - \varphi_o = \Delta\varphi = (n_o - n_e)\cdot d \cdot \frac{2\pi}{\lambda}$$

振幅关系　$A_o = A\sin\alpha$

$A_e = A\cos\alpha$

从晶片出射的是两束传播方向相同、振动方向相互垂直、频率相等、相位差一定的线偏振光，它们合成为一束椭圆偏振光。

2. 波（晶）片

(1) 四分之一波片

$$|n_e - n_o| \cdot d = \frac{\lambda}{4} \rightarrow |\Delta\varphi| = \frac{\pi}{2}$$

$\alpha = \frac{\pi}{4}$ —— 线偏振光→圆偏振光

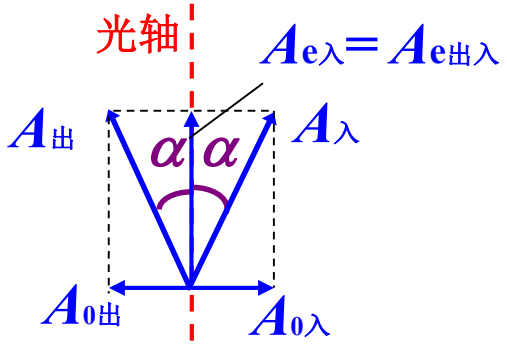
$\alpha = 0, \frac{\pi}{2}$ —— 线偏振光→线偏振光

$\alpha \neq 0, \frac{\pi}{4}, \frac{\pi}{2}$ ——线偏振光→ 正椭圆偏振光

椭圆或圆偏振光经过1/4波片可以获得线偏振光

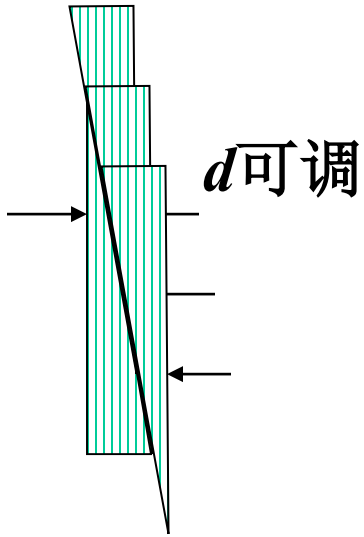
(2) 二分之一波片

$$|n_e - n_o| \cdot d = \frac{\lambda}{2} \rightarrow |\Delta\varphi| = \pi$$



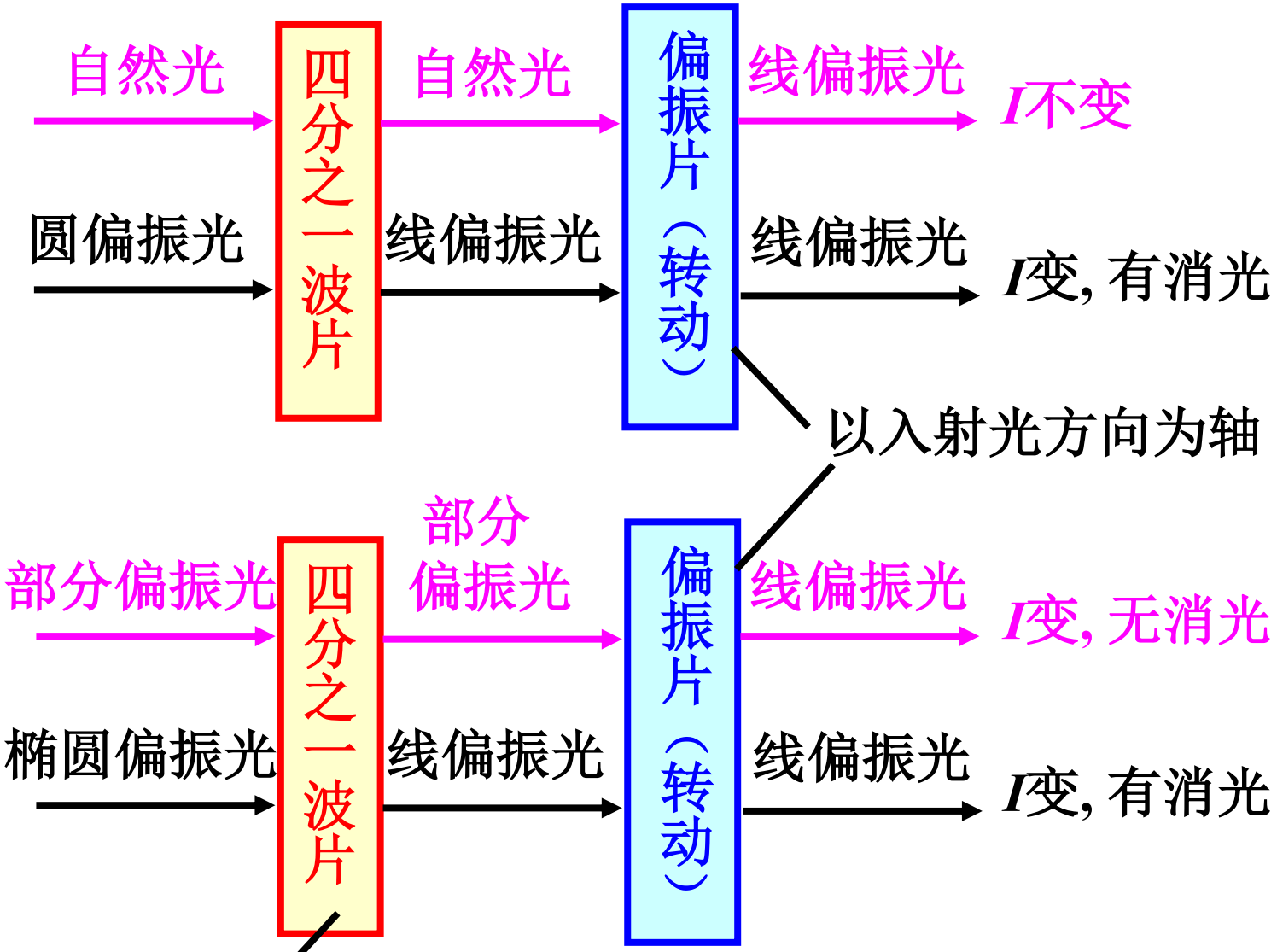
使线偏振光振动面转过一个角度

(3) 可调波片 $|n_o - n_e| d$



二. 椭圆与圆偏振光的检偏

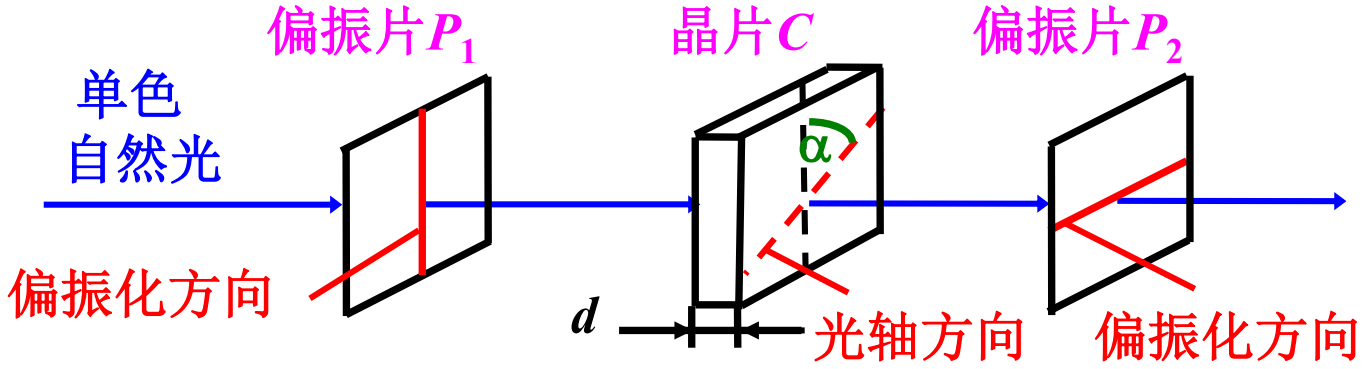
用四分之一波片和偏振片P可区分出自然光和圆偏振光或部分偏振光和椭圆偏振光



光轴平行最大光强或最小光强方向放置或
光轴平行椭圆偏振光的长轴或短轴放置

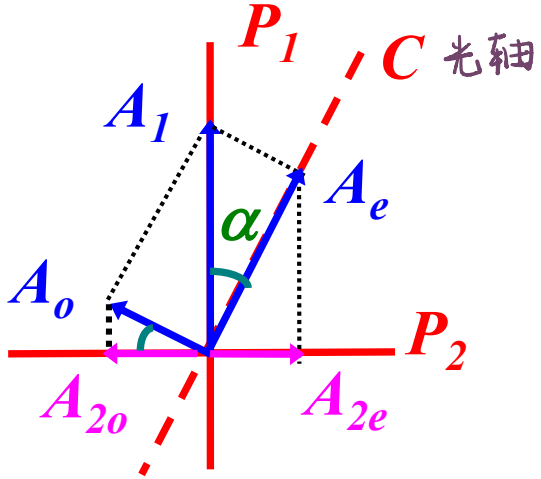
§6　偏振光的干涉

一．偏振光干涉装置



二．偏振光干涉的分析

1．振幅关系



$$A_o = A_1 \sin\alpha$$

$$A_e = A_1 \cos\alpha$$

在P_2后，两束光振动方向平行，振幅为：

$$A_{2o} = A_o \cos\alpha = A_1 \sin\alpha \cdot \cos\alpha$$
$$A_{2e} = A_e \sin\alpha = A_1 \sin\alpha \cdot \cos\alpha = A_{2o}$$

2. 相位关系

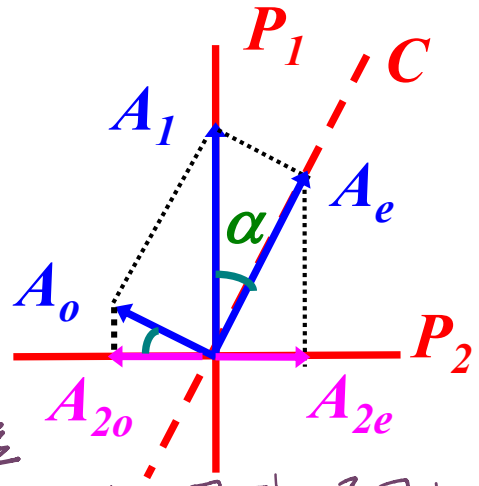
通过晶体C后 $$|\Delta\varphi_c| = \frac{2\pi d}{\lambda}|n_e - n_o|$$

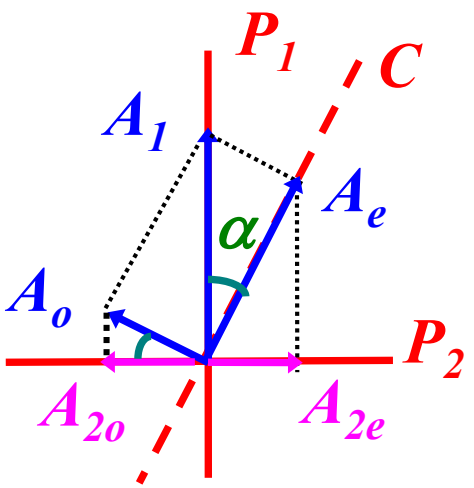
此两束光合成为一束椭圆、圆或线偏振光

通过P_2后

$$|\Delta\varphi| = |\Delta\varphi_c| + \pi = \frac{2\pi d}{\lambda}|n_e - n_o| \underline{+\pi}$$

取决于P1-P2放置
放在光轴两边：+π　光轴同侧：不用加





（若P_1、P_2夹角小于α，则无附加相差π）

$$\Delta\varphi = 2k\pi \rightarrow d = \frac{2k-1}{|n_e - n_o|}\cdot\frac{\lambda}{2},\ (k = 1,2,\cdots)$$

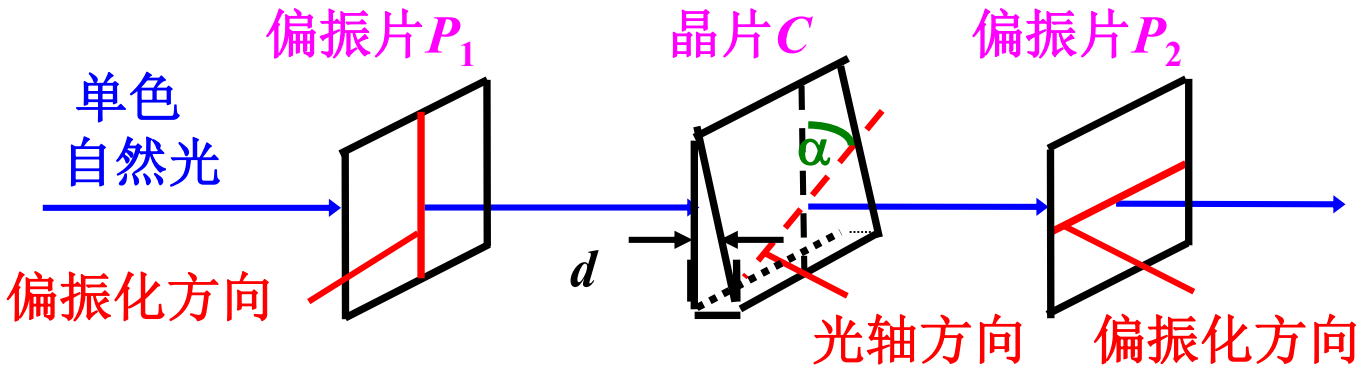
—相长干涉

$$\Delta\varphi = (2k+1)\pi \rightarrow d = \frac{k}{|n_e - n_o|}\lambda$$

—相消干涉

•发生相长或相消干涉时，通过晶片后是线偏振光

若单色光入射，且d不均匀，则屏上出现等厚干涉条纹。

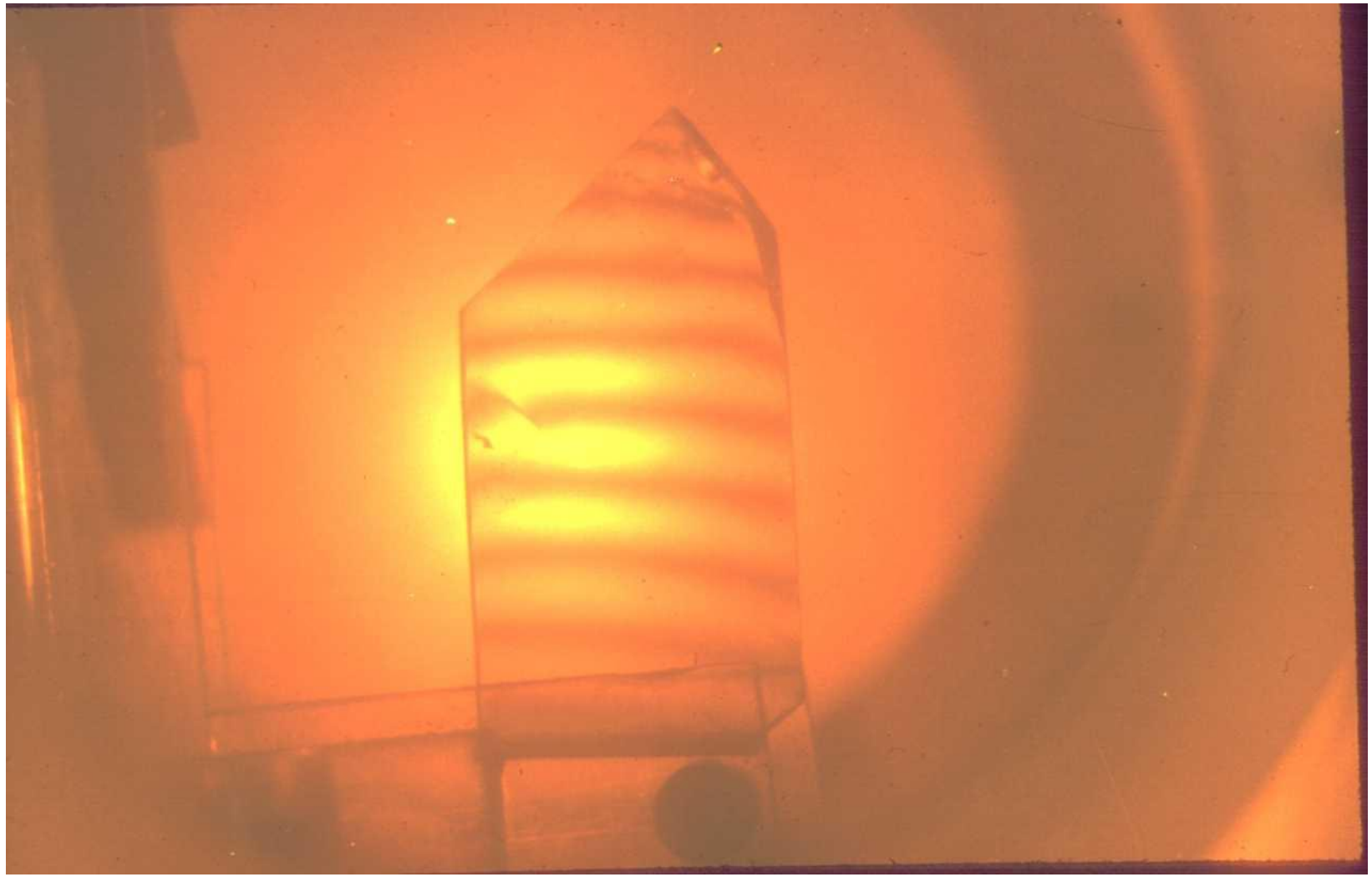


亮纹

$$\Delta\varphi = 2k\pi \rightarrow d = \frac{2k-1}{|n_e - n_o|}\cdot\frac{\lambda}{2},\ (k = 1,2,\cdots)$$

暗纹

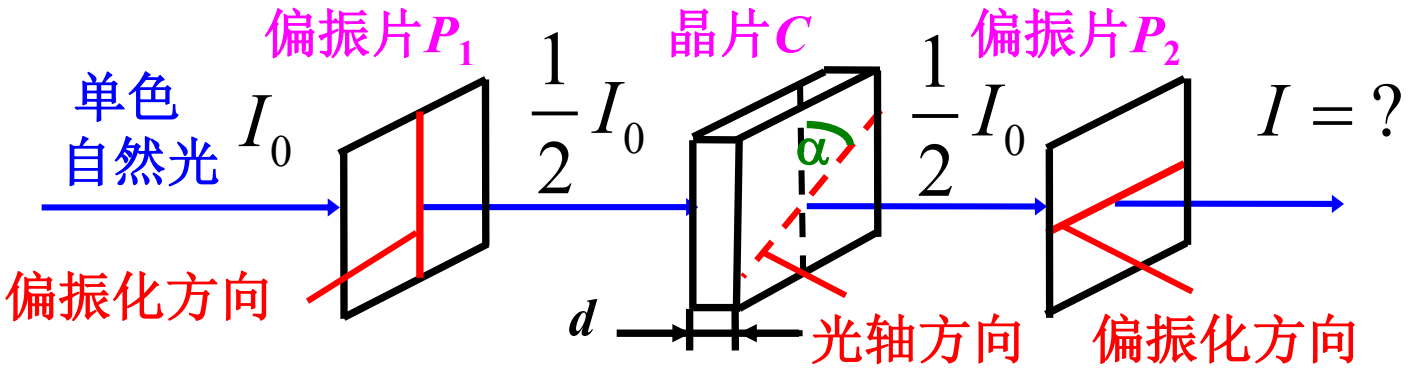
$$\Delta\varphi = (2k+1)\pi \rightarrow d = \frac{k}{|n_e - n_o|}\lambda$$

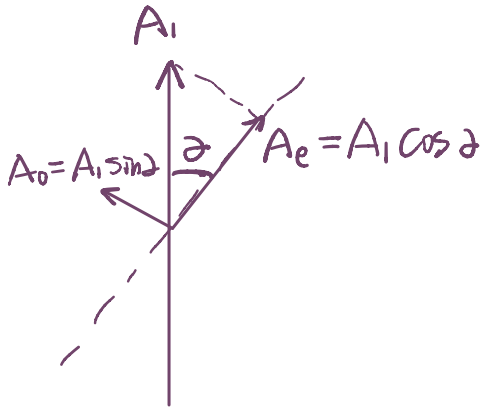


石英劈尖的偏振光干涉（等厚条纹）

例1: 计算各处光强

经过晶片光强不变：不吸收、反射、…
所有光都通过





$$I = A_{2o}^2 + A_{2e}^2 + 2A_{2o}A_{2e}\cos\Delta\varphi$$

$$A_{2o} = A_1 \sin\alpha \cdot \cos\alpha = A_{2e}$$

$$\frac{1}{2}I_0 = A_1^2$$

$$I = I_0 \cos^2\alpha \sin^2\alpha (1+\cos\Delta\varphi)$$

$$|\Delta\varphi| = |\Delta\varphi_c| + \pi = \frac{2\pi d}{\lambda}|n_e - n_o| + \pi$$

三. 色偏振

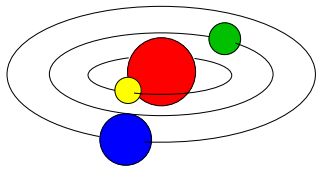
白光入射，晶片 d 均匀，不同波长对应不同相位差

$$|\Delta\varphi| = \frac{2\pi d}{\lambda}|n_e - n_o| + \pi$$

屏上由于某种颜色干涉相消，而呈现它的互补色这叫（显）色偏振。

如　红色相消→青绿色；蓝色相消→红黄色

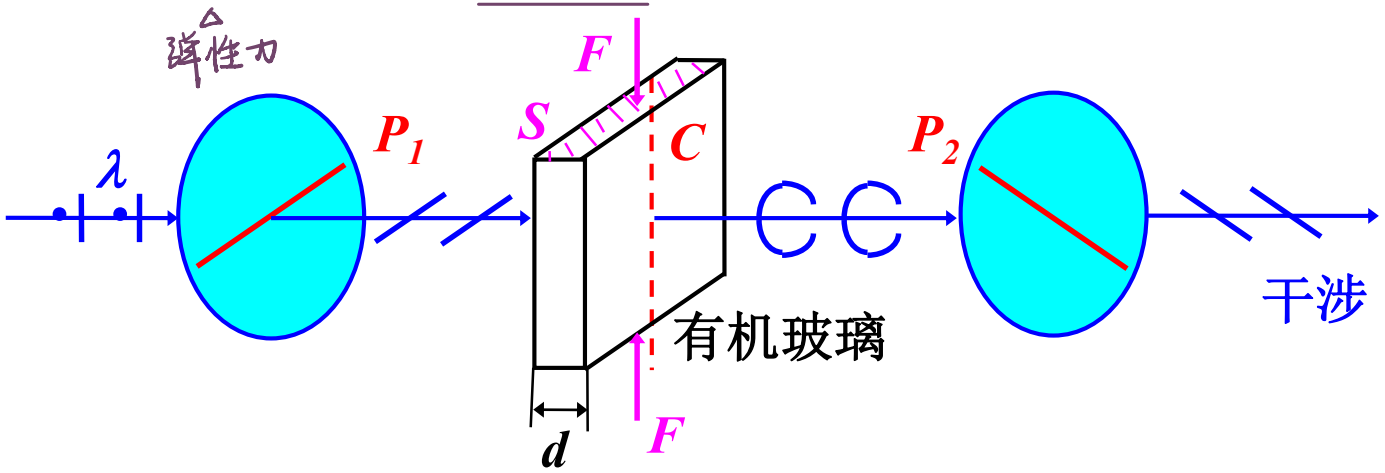
若 d 不均匀，则屏上出现彩色条纹。



§7　人工双折射

人为地造成各向异性，而产生双折射。

一、光弹效应(应力双折射效应)



应力→各向异性→υ各向不同→n各向不同

在一定应力范围内：

应力↑ 条纹密集↑

$$|n_e - n_o| = k\frac{F}{S}$$

$$|\Delta\varphi| = \frac{2\pi d}{\lambda}|n_e - n_o| + \pi = \frac{k \cdot d \cdot 2\pi}{\lambda} \cdot \frac{F}{S} + \pi$$

各处 $\frac{F}{S}$ 不同→各处 $\Delta\varphi$ 不同→出现干涉条纹

$\frac{F}{S}$ 变→ $\Delta\varphi$ 变→干涉情况变。

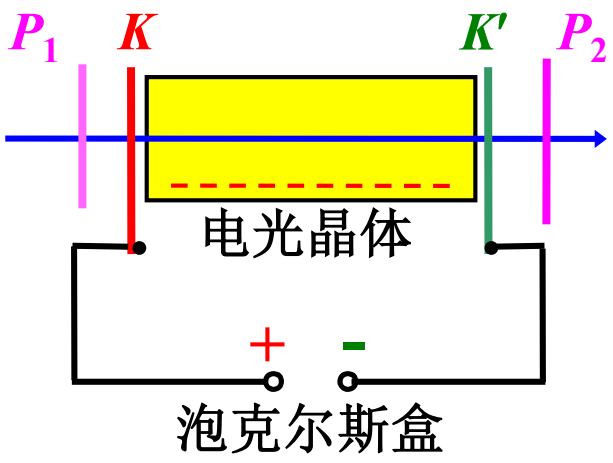
二. 电光效应

克尔效应：加电场→液体呈单轴晶体性质

科顿—穆顿效应：

加磁场→液体呈单轴晶体性质

泡克尔斯效应(1893年)



不加电场→P_2不透光

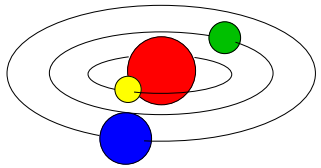
加电场→晶体变双轴晶体

原光轴方向附加了双折射效应→P_2透光

泡克尔斯效应引起的相位差 $\Delta\varphi \propto V$

应用： 光开关(响应时间很短)

nm量级

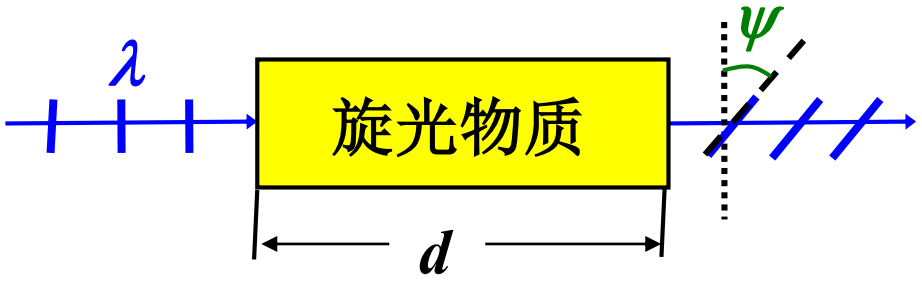


§8 旋光现象　Roto-Optical Phenomena

一. 物质的旋光性

使线偏振光的振动面发生旋转

左旋物质
右旋物质



旋转的角度：$\boxed{\psi = a \cdot d}$　a — 旋光率 $\sim \lambda^{-2}$

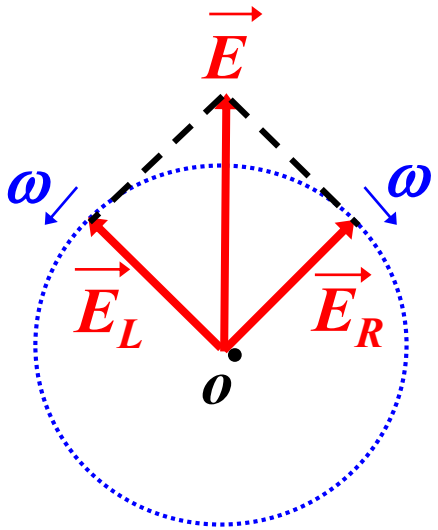
二. 菲涅耳的解释

旋光色散

不同波长 经过相同长度的旋光物质 转过角度不同

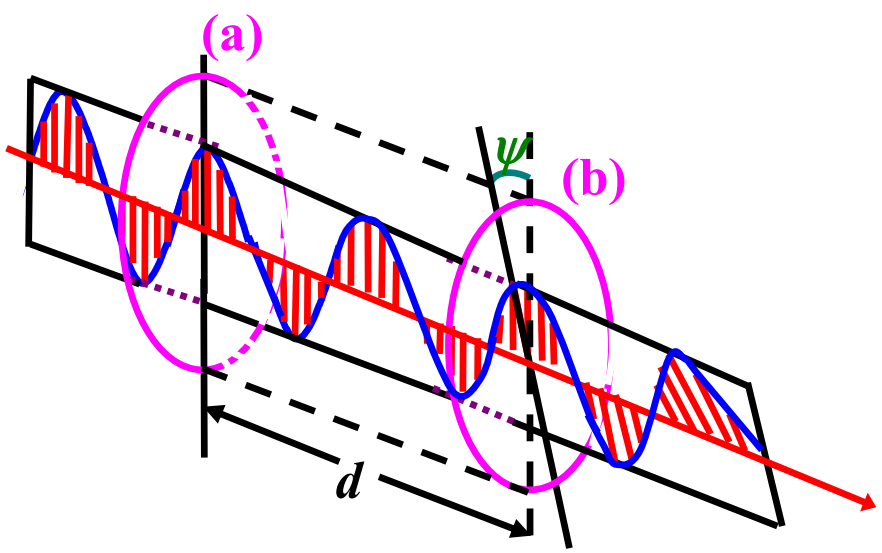
线偏振光可看作是

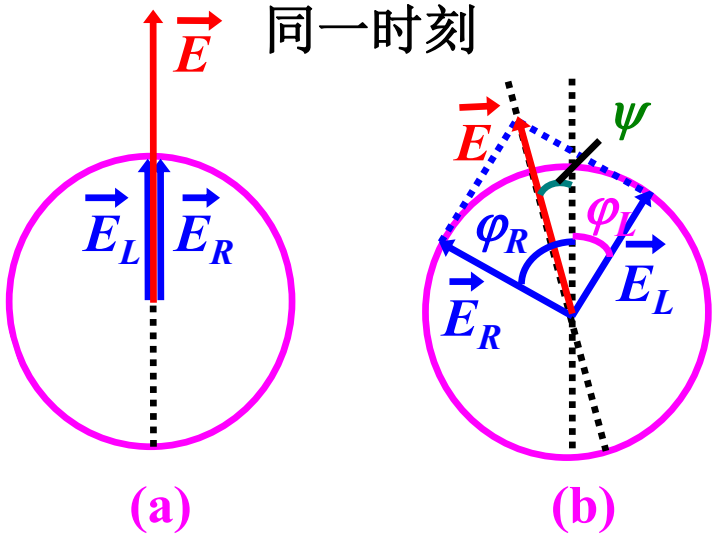
同频率、等振幅、有确定相位差的左（L ）、右（$\boldsymbol{R}$ ）旋圆偏振光的合成。



旋光物质 $\upsilon_L \neq \upsilon_R \rightarrow n_L \neq n_R$

偏振光通过旋光物质后，
初相滞后有差异





设入射时L，R初相为0，旋光物质长d

$n_R > n_L$（即$\upsilon_L > \upsilon_R$）

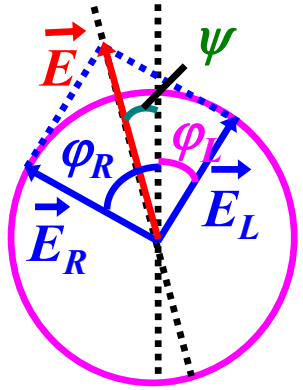
振动面逆时针 ↺

左旋物质：$v_L > v_R$，

$n_R > n_L$

在出射面上:

$$\varphi_R = -\frac{n_R d}{\lambda}\cdot 2\pi < 0 \qquad \varphi_L = -\frac{n_L d}{\lambda}\cdot 2\pi < 0$$



$$|\varphi_R| > |\varphi_L|$$

由图示，有:

$$\psi = \frac{1}{2}[|\varphi_R| + |\varphi_L|] - |\varphi_L| = \frac{1}{2}[|\varphi_R| - |\varphi_L|]$$

$$= \frac{-1}{2}[\varphi_R - \varphi_L] = \frac{\pi}{\lambda}(n_R - n_L)\cdot d$$

令 $a = \frac{\pi}{\lambda}(n_R - n_L)$ —— 旋光率

则 $\psi = a \cdot d$

三. 浓度测量

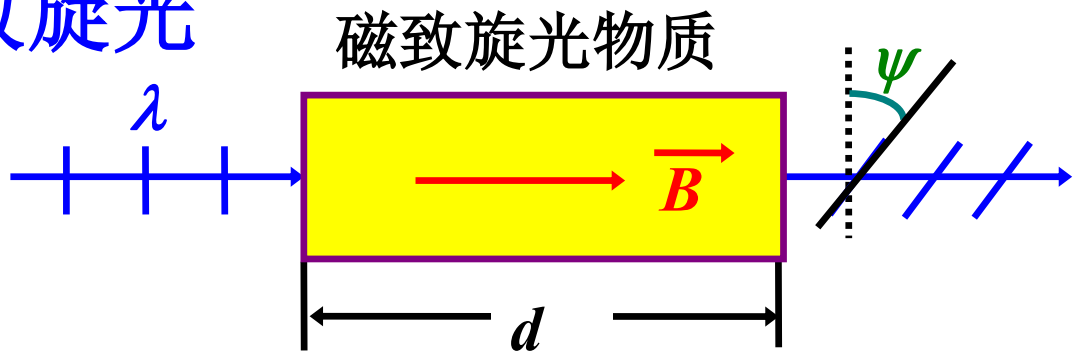
对溶液有 $\psi = a' \cdot c \cdot d$

$a' \cdot c = a$ —— 溶液的旋光率

c —— 溶液的浓度

a' —— 溶液的比旋光率

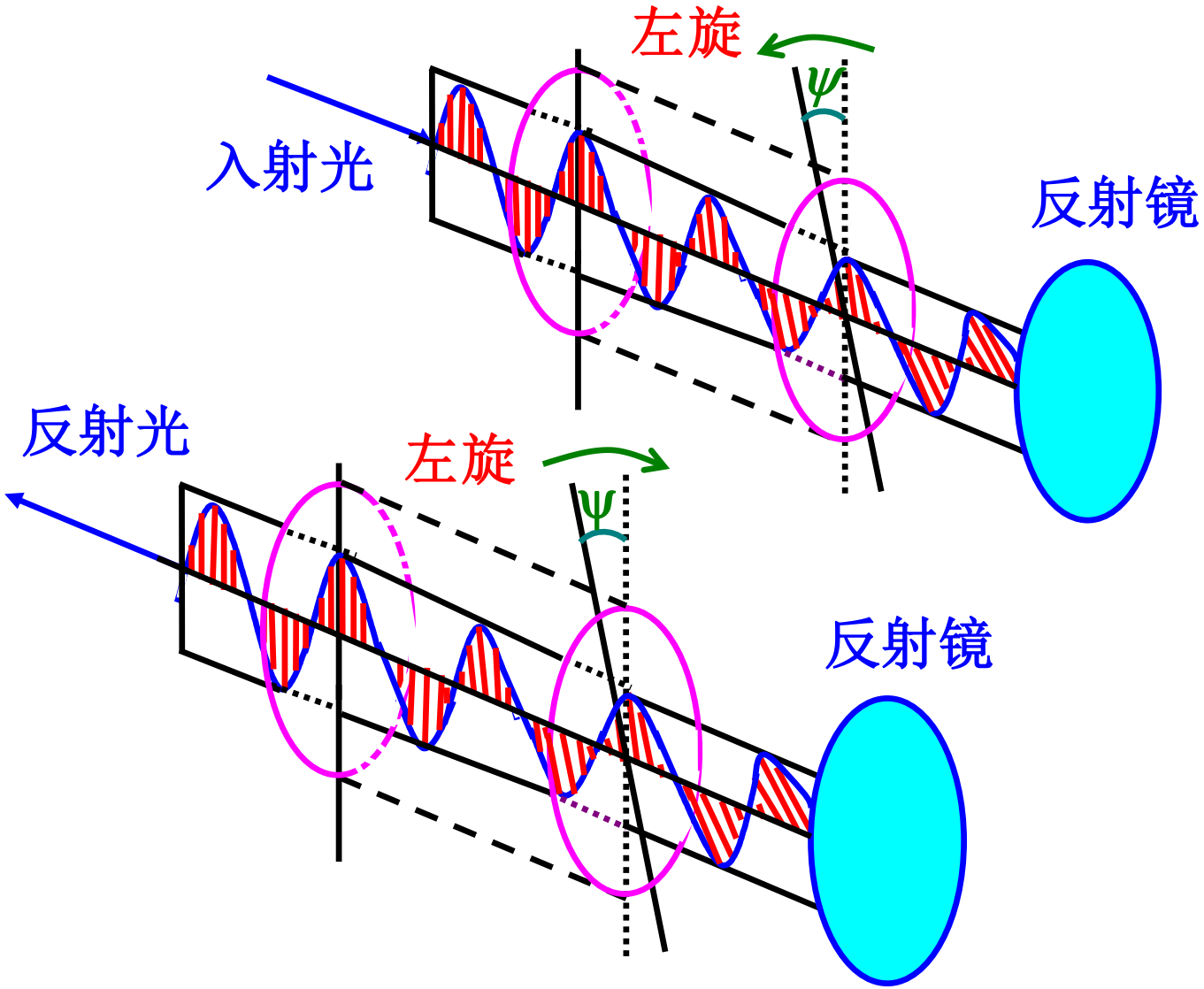
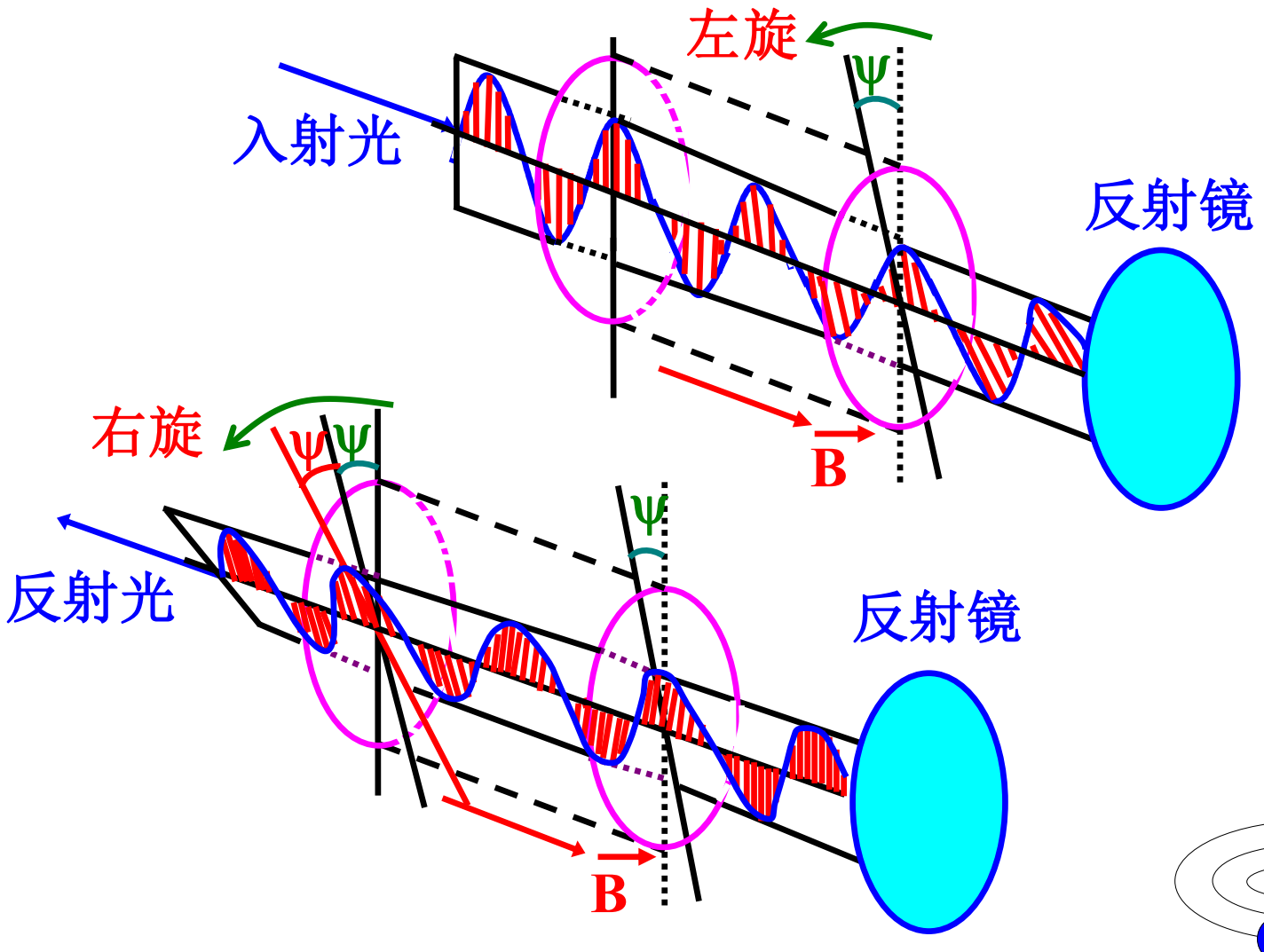
四、磁致旋光



旋转的角度 $\psi = V \cdot d \cdot B$

V—费德尔常量 $\sim 10^4 - 10^5\ \mathrm{m}^{-1} \cdot \mathrm{T}^{-1}$

- 对自然旋光物质，振动面的左旋或右旋是由旋光物质本身决定的，与光的传播方向无关。



- 对磁致旋光物质，光沿 $\vec{B}$ 与逆 $\vec{B}$ 方向传播，振动面旋向相反。